Splunk Alert Creation and Validation

Objective –

The objective of this phase was to create real-time security alerts in Splunk for critical Windows security events and validate that the alerts were triggered correctly when the corresponding activities occurred on the monitored Windows endpoint.

➔ Alert 1: New User Account Creation Detection

Purpose -

User account creation is a common administrative activity but can also indicate unauthorized persistence by an attacker. Monitoring newly created accounts helps security teams identify suspicious account creation attempts.

Detection Logic -

Windows Security Event ID:

- Event ID 4720 – A user account was created

Splunk Search -

- EventCode=4720

Alert Configuration -

- Alert Name: New User Created
- Alert Type: Real-Time
- Trigger Condition: Per Result
- Severity: High

Validation Procedure -

A new local user account was created using:

- net user demo /add

The Windows Security log generated Event ID 4720.

Result -

The alert was successfully triggered and appeared in Splunk Triggered Alerts.

➔ Alert 2: User Account Deletion Detection

Purpose -

User account deletion events can indicate administrative cleanup or malicious attempts to remove accounts and cover tracks. Monitoring these events improves account management visibility.

Detection Logic -

Windows Security Event ID:

- Event ID 4726 – A user account was deleted

Splunk Search -

- EventCode=4726

Alert Configuration -

- Alert Name: User Account Deletion
- Alert Type: Real-Time
- Trigger Condition: Per Result
- Severity: Medium

Validation Procedure -

The previously created account was deleted using:

net user demo /delete

Windows generated Event ID 4726.

Result -

The alert was successfully triggered and appeared in the Triggered Alerts dashboard.

➔ Alert 3: Remote Desktop Protocol (RDP) Login Detection

Purpose -

RDP is commonly used by administrators for remote management. However, attackers frequently use RDP for lateral movement and unauthorized access. Monitoring RDP logins is essential for detecting suspicious remote access activity.

Detection Logic -

Windows Security Event ID:

- Event ID 4624
- Logon Type 10 (Remote Interactive / RDP)

Splunk Search -

EventCode=4624 Logon_Type=10

Alert Configuration -

- Alert Name: RDP Login Detection
- Alert Type: Real-Time
- Trigger Condition: Per Result
- Severity: High

Validation Procedure -

An RDP session was initiated from the Parrot Security VM to the Windows 10 target using FreeRDP.

Successful authentication generated an Event ID 4624 with Logon Type 10.

Result -

The alert successfully triggered and appeared in the Triggered Alerts dashboard.

➔ Alert 4: Failed Login Attempt Detection

Purpose -

Failed login attempts may indicate unauthorized access attempts, password guessing attacks, brute-force attacks, or credential stuffing activities. Monitoring failed authentication events enables security teams to identify suspicious login behavior and investigate potential threats before successful compromise occurs.

Detection Logic -

Windows Security Event ID:

- Event ID 4625 – An account failed to log on

Splunk Search -

- EventCode=4625

Alert Configuration -

- Alert Name: Failed Login Detection
- Alert Type: Real-Time
- Trigger Condition: Per Result
- Severity: Medium

Validation Procedure -

A failed login attempt was intentionally generated by entering an incorrect password during Windows authentication.

The Windows Security Log recorded Event ID 4625 indicating that the authentication request was unsuccessful.

Result -

The alert was successfully triggered and appeared in the Splunk Triggered Alerts dashboard.

The generated event contained information such as:

- Account Name
- Failure Reason
- Logon Type
- Source Workstation
- Timestamp

This information can assist SOC analysts in identifying and investigating suspicious authentication attempts.

Security Impact

The implemented alerts provide visibility into critical account management and remote access activities, including:

- New account creation
- User account deletion
- Remote Desktop logins
- Unauthorized access attempt

These detections improve monitoring capabilities and demonstrate the practical implementation of a Security Operations Center (SOC) workflow using Splunk Enterprise.

Conclusion

The Splunk SOC Home Lab successfully demonstrated real-time detection and alerting for critical Windows security events. Security alerts were configured, validated, and triggered using controlled attack simulations and administrative actions. The lab provides foundational experience in log monitoring, threat detection, alert creation, and SOC operations.

Scenario 1 – User Account Creation (Event ID 4720)

Figure 1: User Account Creation Command Execution

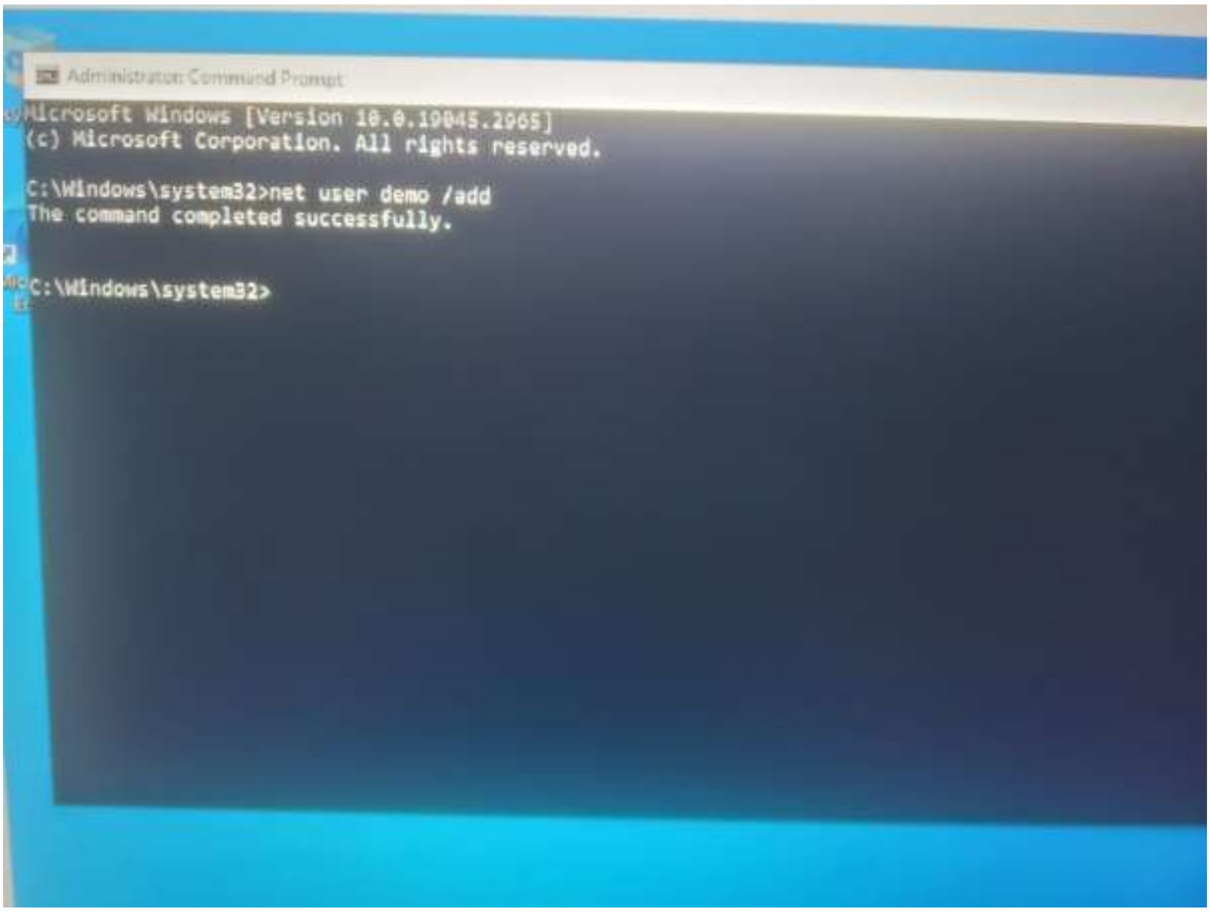


Figure 2: Windows Event Viewer Showing Event ID 4720

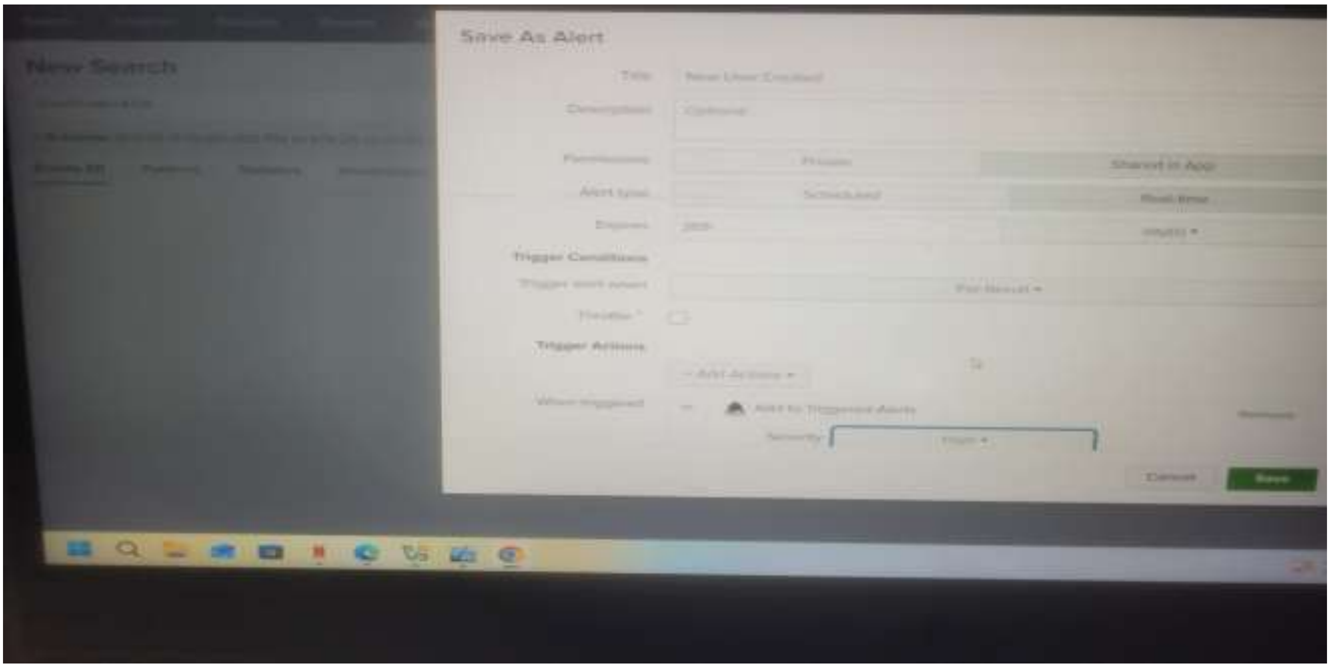
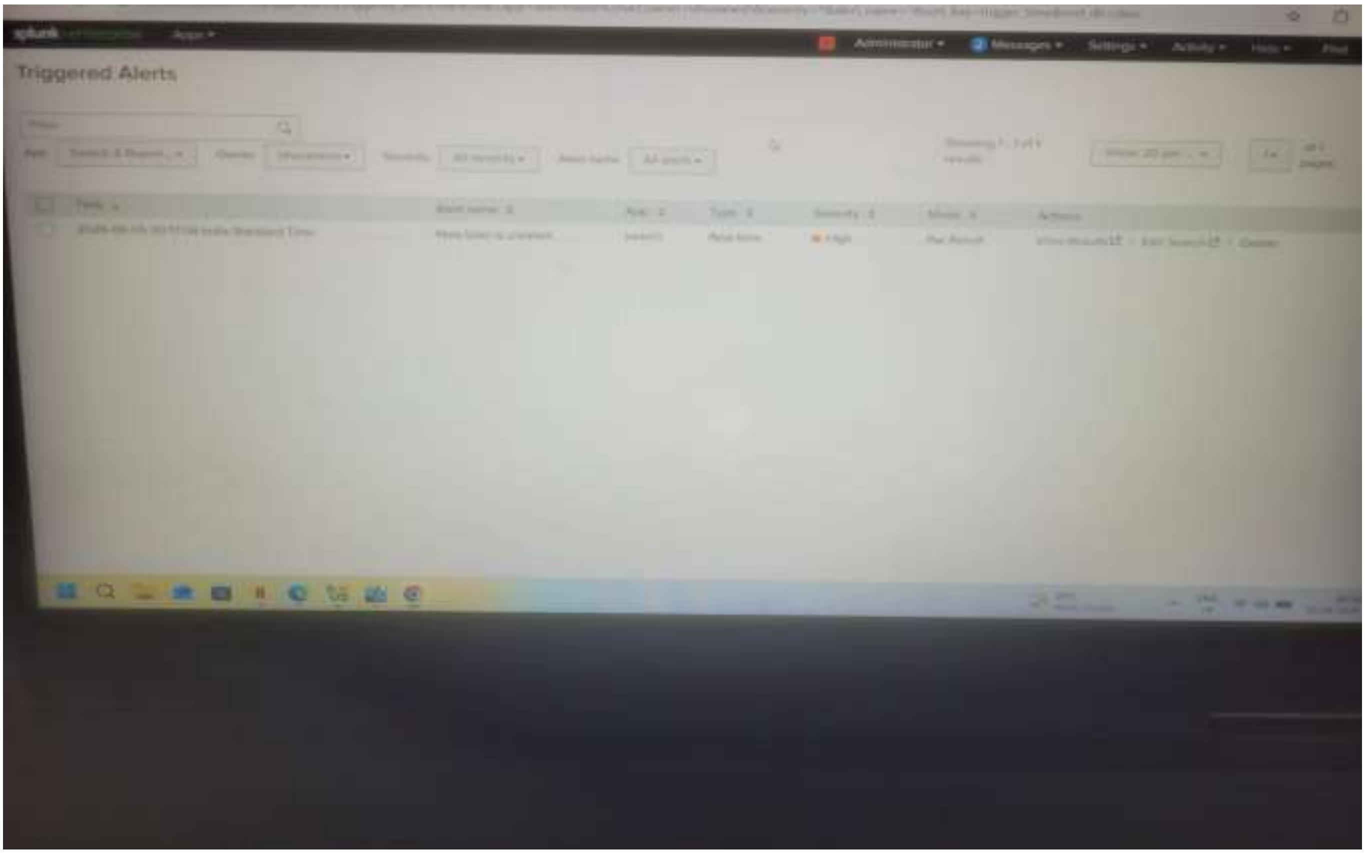


Figure 3: Splunk Detection of User Account Creation Event (EventCode=4720)



Scenario 2 – User Account Deletion (Event ID 4726)

Figure 4: User Account Deletion Command Execution

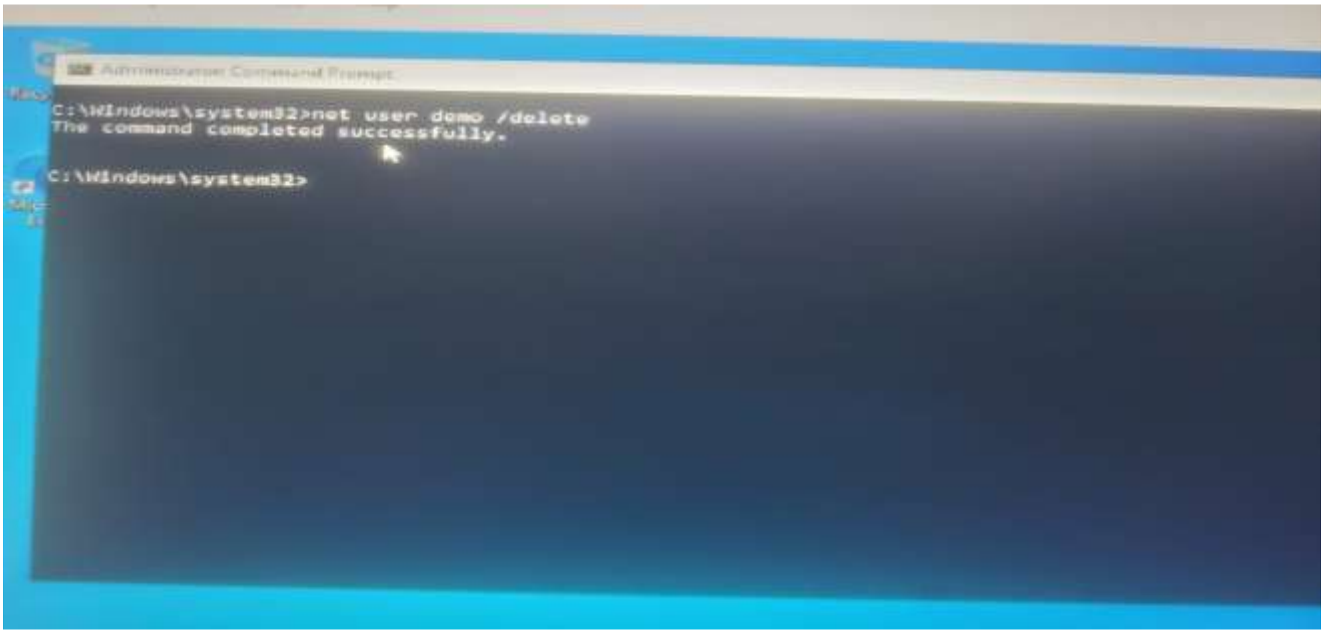
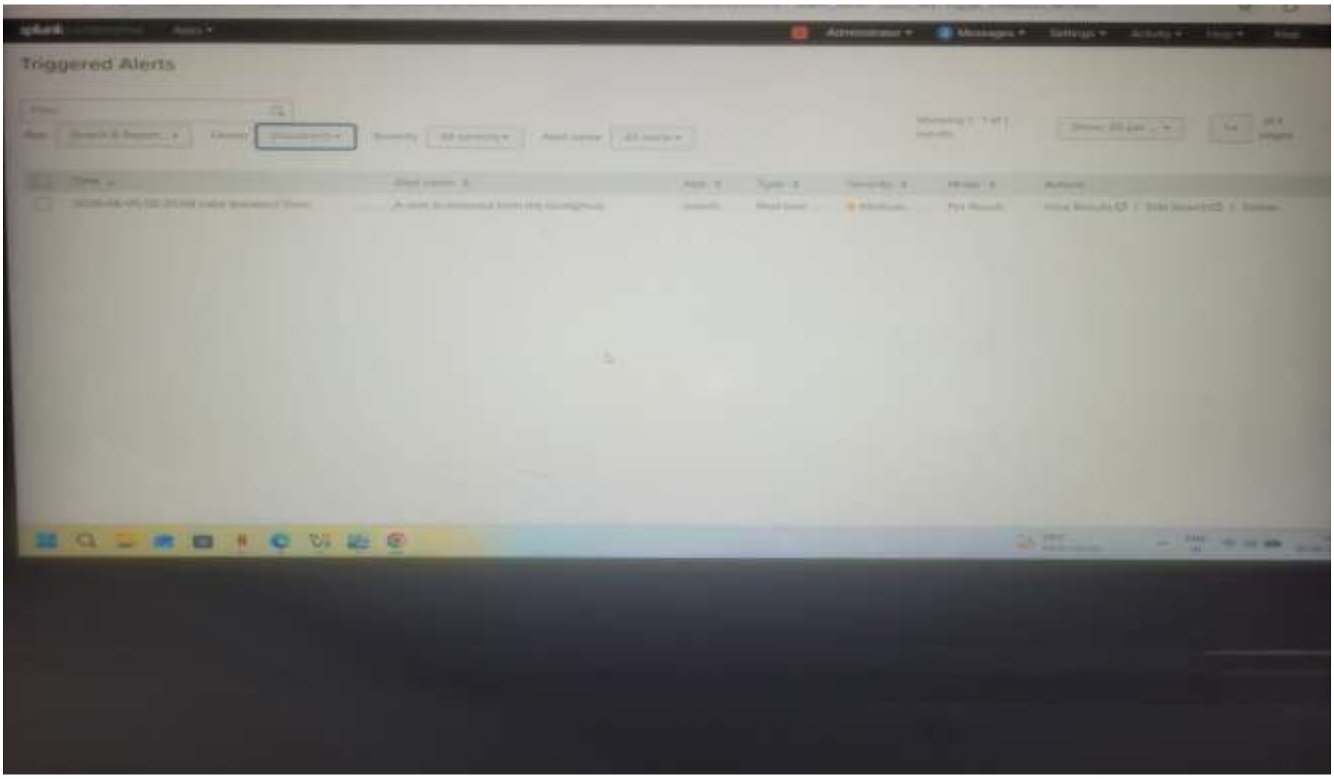


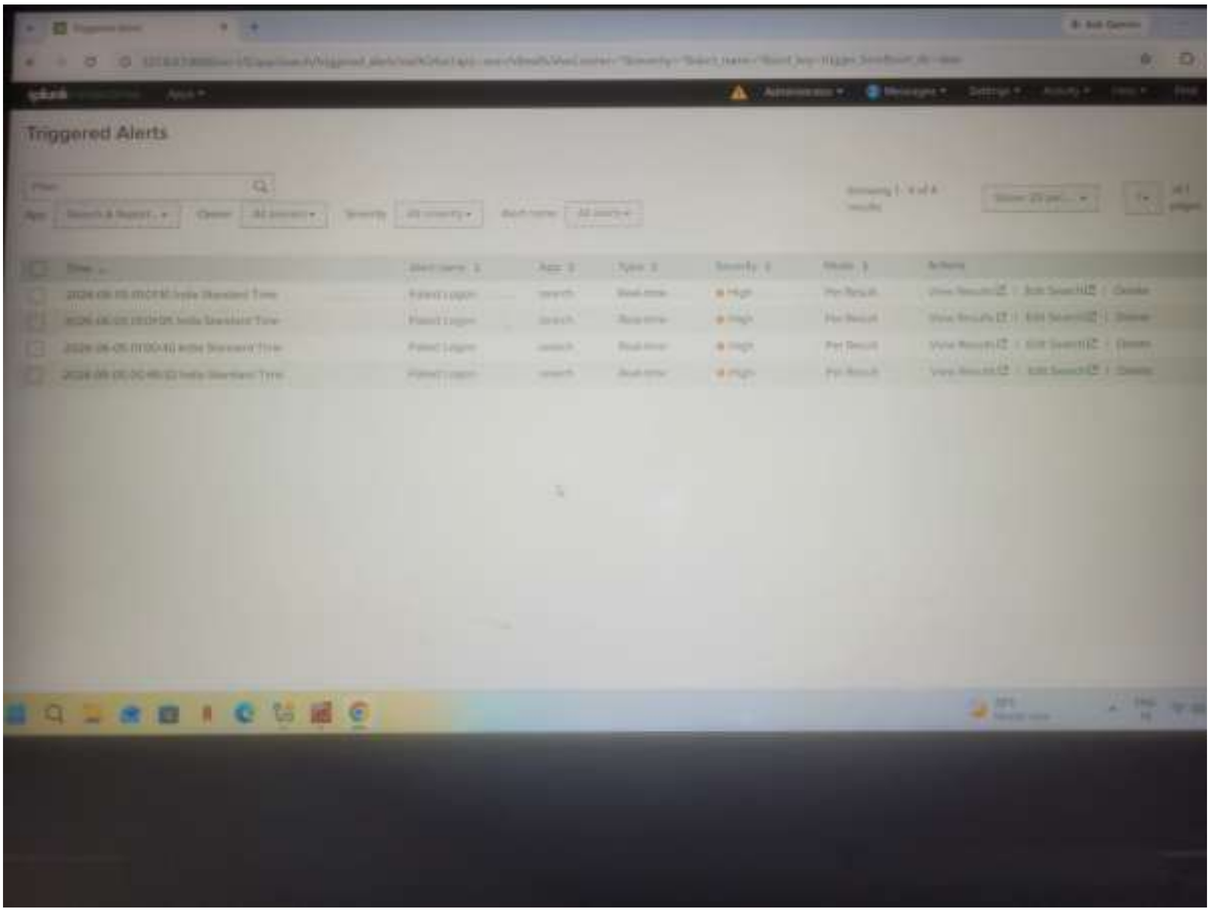
Figure 5: Splunk Detection of User Account Deletion Event (EventCode=4726)



Scenario 3 – Failed Login Detection (Event ID 4625)

Figure 6: Failed Login Attempt on Windows System

Figure 7: Splunk Detection of Failed Login Event (EventCode=4625)



Scenario 4 – RDP Login Detection

Figure 8: RDP Connection Initiated from Parrot OS

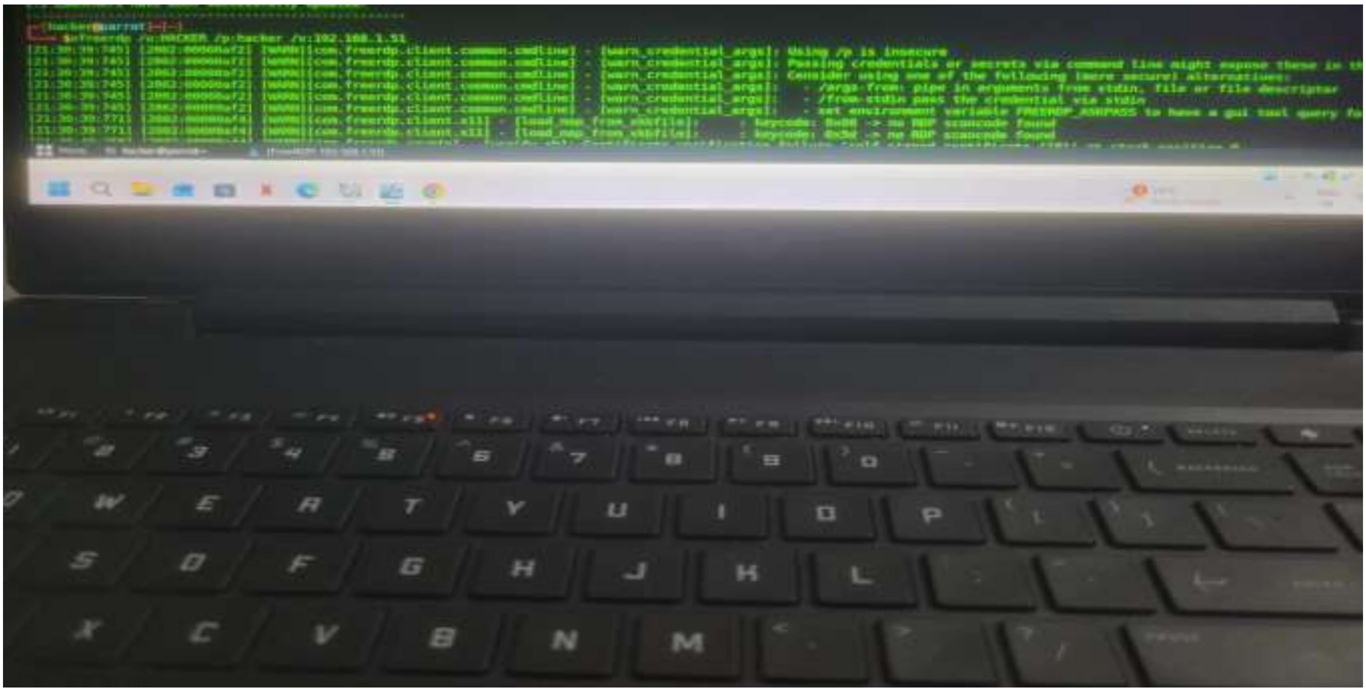


Figure 9: Successful Remote Desktop Session to Windows Target

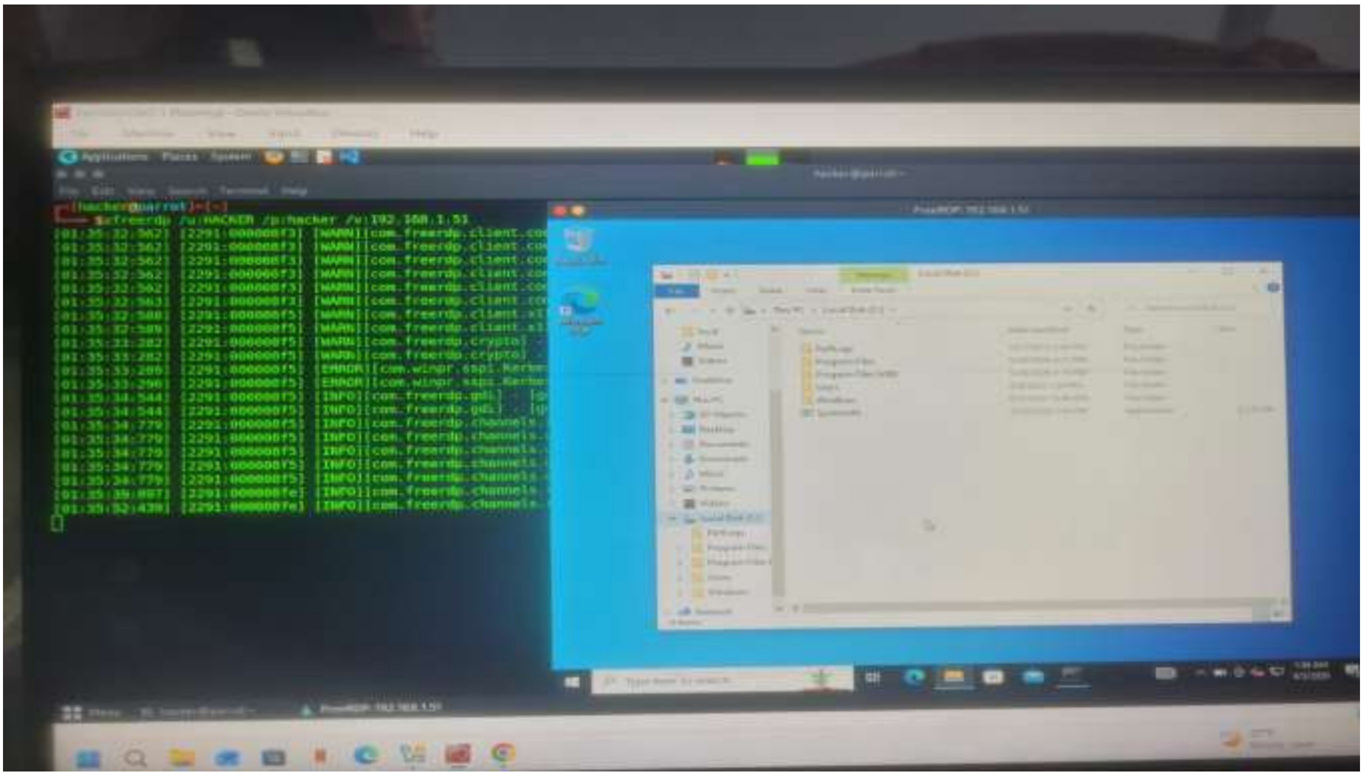


Figure 10: Splunk Detection of RDP Login Event